Every great change starts with a clear vision.

Arnold Schwarzenegger

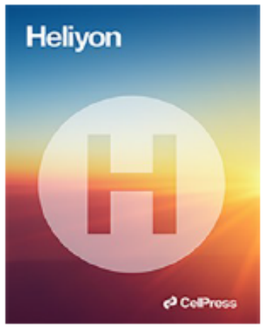
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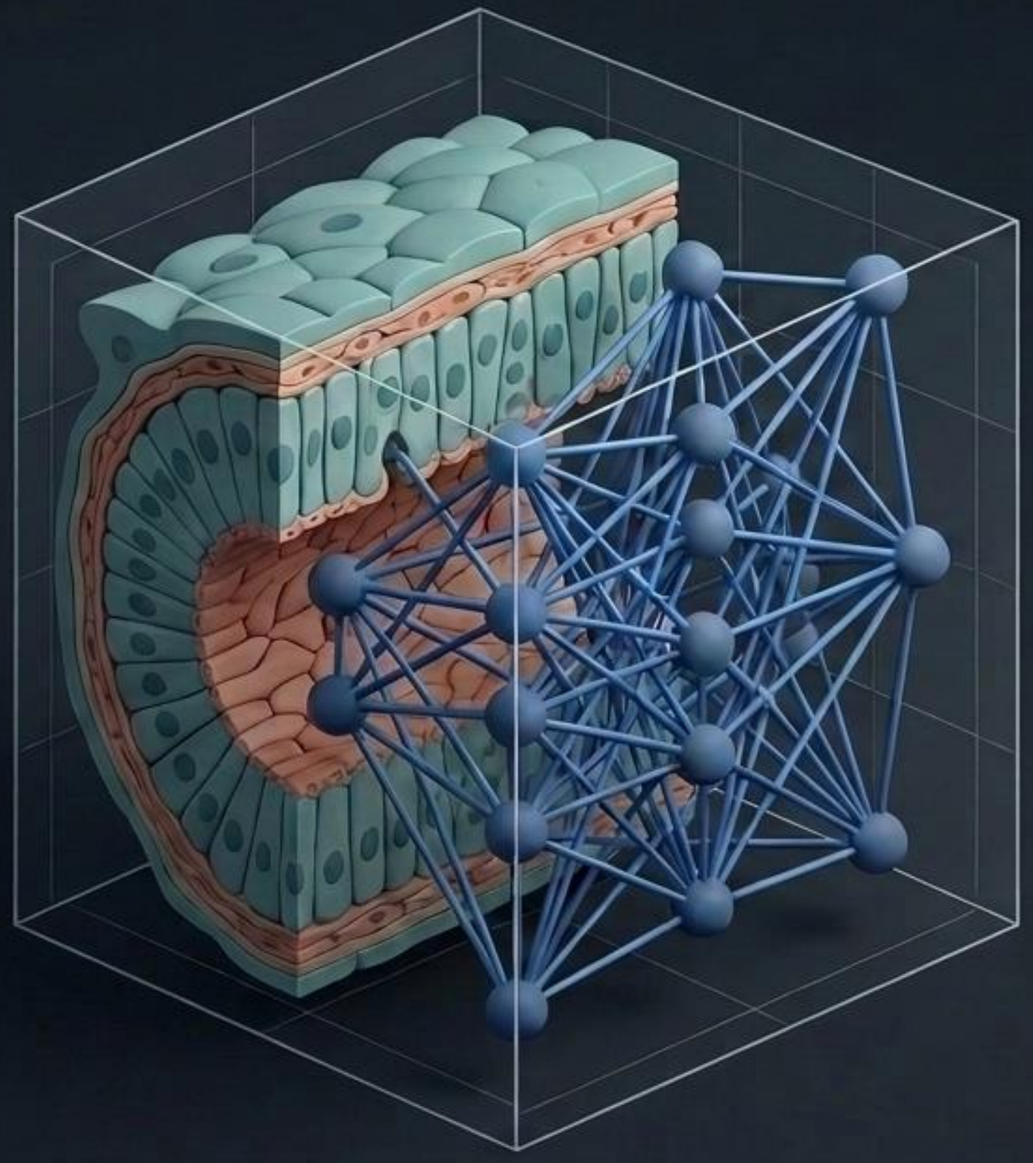
journal homepage: www.cell.com/heliyon



A deep learning-based framework for predicting survival-associated groups in colon cancer by integrating multi-omics and clinical data

Siamak Salimy [a], Hossein Lanjanian [b], Karim Abbasi [c], Mahdieh Salimi [d], Ali Najafi [e], Leili Tapak [f], Ali Masoudi-Nejad [a,*,1]

[a] *Laboratory of System Biology and Bioinformatics (LBB), Department of Bioinformatics, University of Tehran, Kish International Campus, Kish, Iran*
[b] *Cellular and Molecular Endocrine Research Center, Research Institute for Endocrine Sciences, Shahid Beheshti University of Medical Sciences, Tehran, Iran*
[c] *Laboratory of System Biology, Bioinformatics & Artificial Intelligent in Medicine (LBBai), Faculty of Mathematics and Computer Science, Kharazmi University, Tehran, Iran*
[d] *Department of Medical Genetics, Institute of Medical Biotechnology, National Institute of Genetic Engineering and Biotechnology (NIGEB), Tehran, Iran*
[e] *Molecular Biology Research Center, Systems Biology and Poisonings Institute, Tehran, Iran*
[f] *Department of Biostatistics, School of Public Health and Modeling of Noncommunicable Diseases Research Center, Hamadan University of Medical Sciences, Hamadan, Iran*



39th Journal Club | Student Branch of the International Society for Computational Biology (ISCB) in Iran

Decoding Colon Cancer Survival: A Deep Learning-Based Framework

Integrating Multi-Omics and Clinical Data via Hyperparameter-Optimized Autoencoders

Authors: Siamak Salimy, Hossein Lanjanian, Karim Abbasi, Mahdieh Salimi, Ali Najafi, Leili Tapak, Ali Masoudi-Nejad

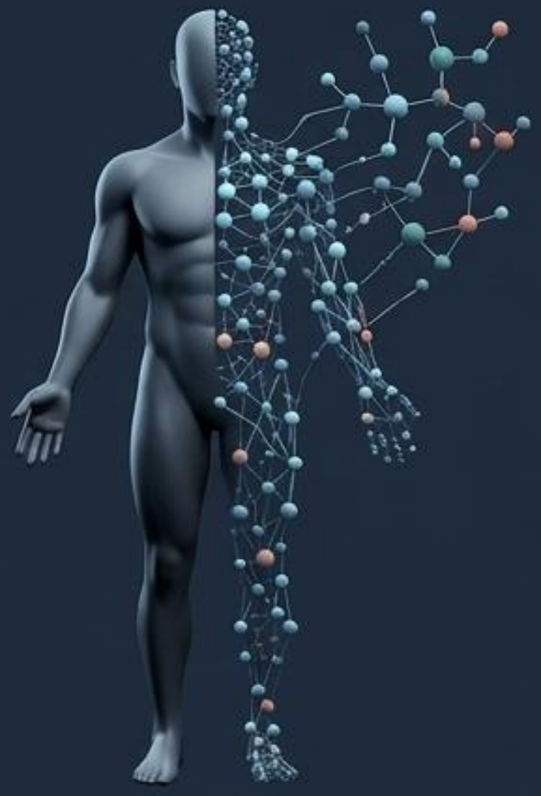
Journal: Heliyon, Vol. 9, e17653 (2023) | DOI: 10.1016/j.heliyon.2023.e17653

Invited Speaker: Dr. Siamak Salimy | Director, BCRLAB

Gemini Notebook

The Clinical Reality & The Computational Bottleneck

1

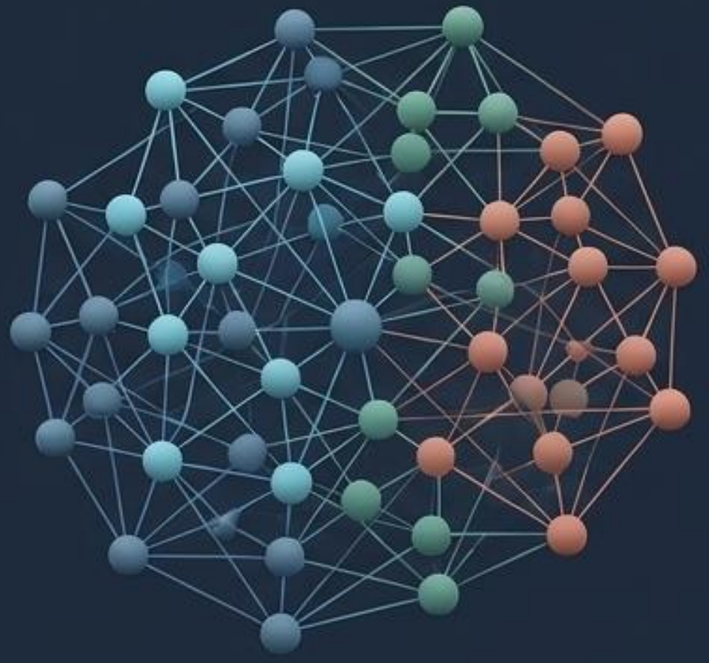


The Staging Limitation

Colorectal cancer 5-year survival rates vary drastically (95% Stage I to 11% Stage IV).

The Gap: Current clinical treatments lack an effective, stage-independent survival classification system.

2



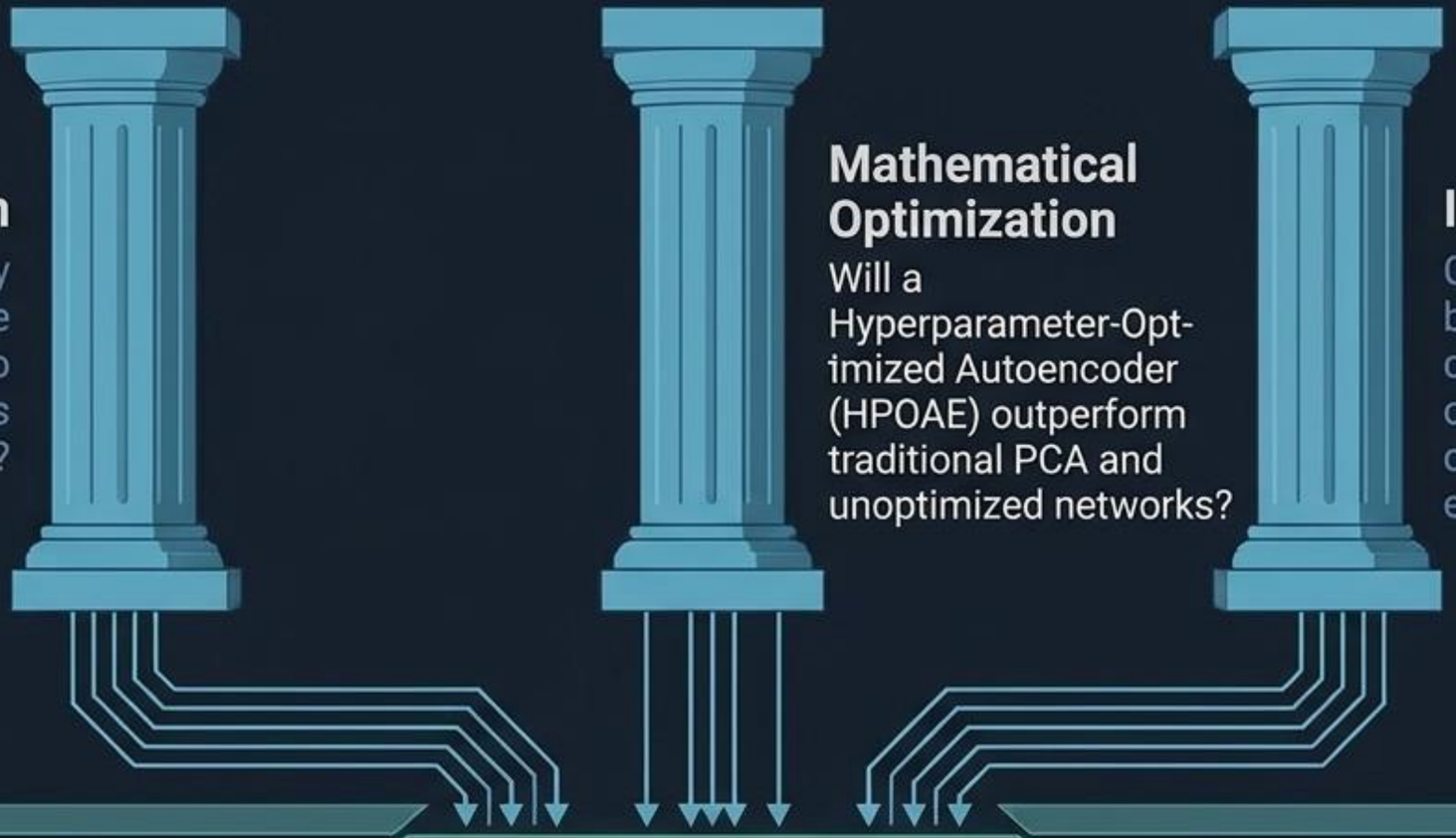
Multi-Omics Chaos

Genomics, transcriptomics, and epigenomics hold vital prognostic signatures.

The Barrier: Fusing massive dimensionality across mRNA, miRNA, and DNA methylation creates severe computational complexity that traditional statistics cannot easily resolve.

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Methodological Rationale and Scientific Objectives



Data Integration

Can we simultaneously integrate three disparate molecular data types to out-predict single-omics models?

Mathematical Optimization

Will a Hyperparameter-Optimized Autoencoder (HPOAE) outperform traditional PCA and unoptimized networks?

Interpretable Rules

Can deep learning's black box be translated into concrete, supervised conditional rules for clinical biomarker extraction?

The Claimed Contribution

A highly precise, deep-learning survival subgroup prediction model capable of distinguishing colon cancer risk independently of clinical stage.

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Study Cohorts: Tracking Sample Availability

Flow 1 (Discovery - TCGA-COAD)

Reported Base Cohort: 368 patients
(Abstract/Methods)

- **mRNA** (267)
- **miRNA** (262)
- **Methylation** (211)

Intersection Used:
211 samples

⚠ Amber Flag - Reporting Discrepancy

While the title claims clinical data integration, the 3-omics overlap used for the deep learning bottleneck was constrained to 211 samples missing clinical integrations in the autoencoder.

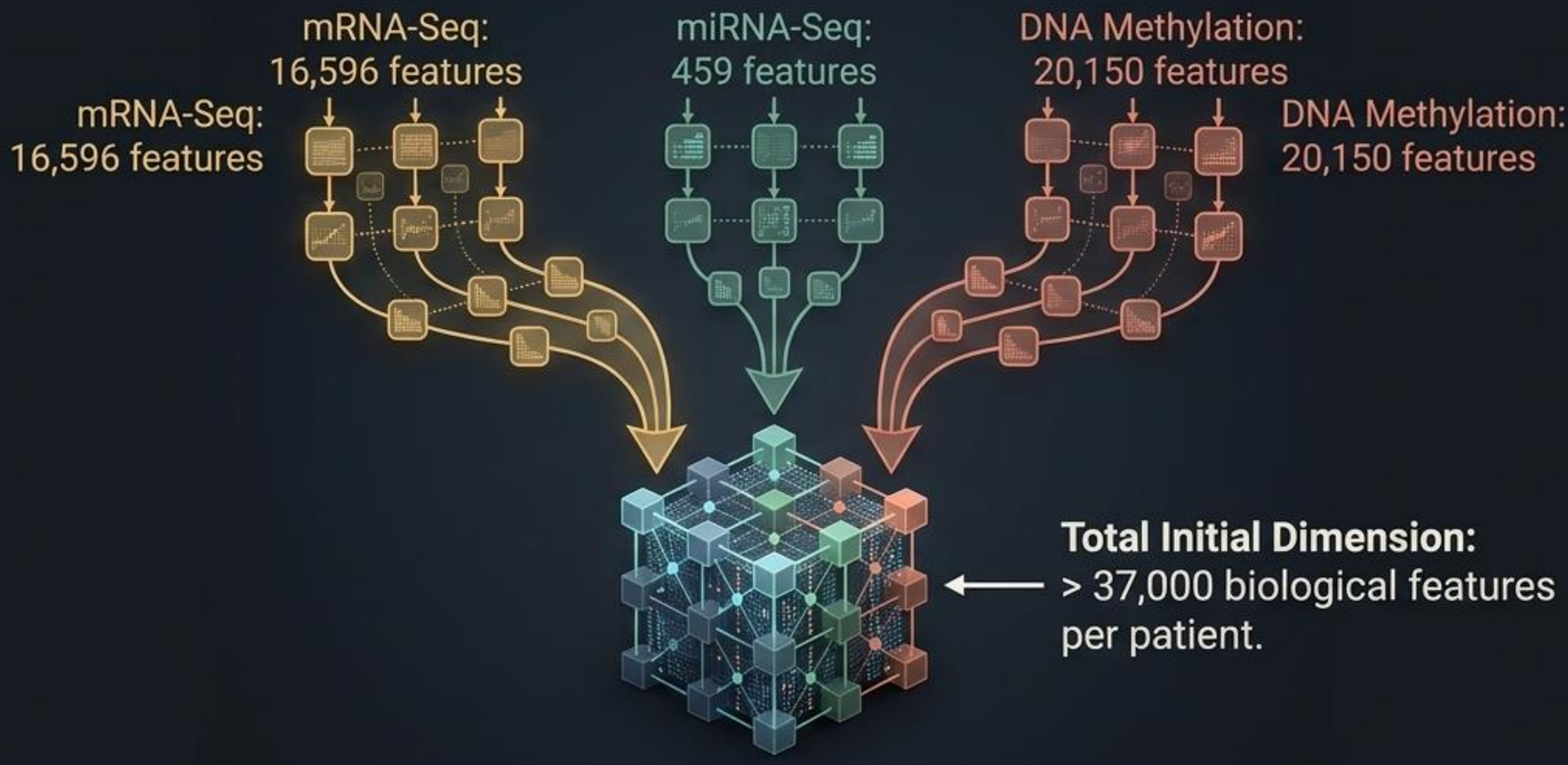
Flow 2 (Independent Confirmation)

E-GEOD-17538 Dataset (232 samples)

Contains RNA-Seq expression data used to validate the model's structural clusters.

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Ingesting the Raw Architecture of Cancer



Total Initial Dimension:
> 37,000 biological features per patient.

Pre-Processing Methodology:

- Matrices were unit-norm scaled.
- Stacked sample-by-sample to create a unified, high-dimensional input matrix prior to dimensionality reduction.

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The Two-Phase Deep Learning Framework

Phase A: Unsupervised Subgroup Discovery

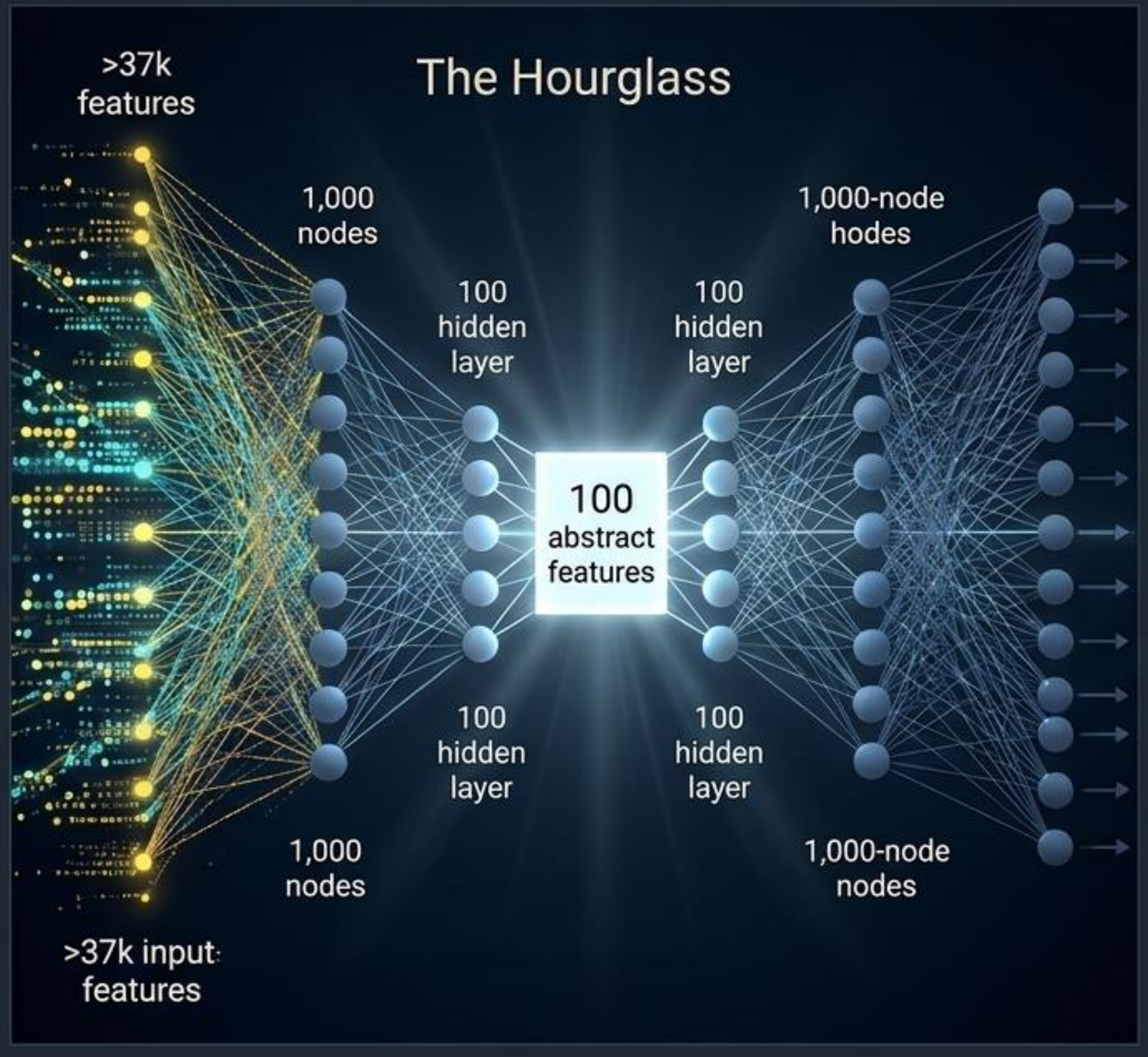
1. Multi-omics integration & scaling.
2. Dimensionality reduction via HPOAE.
3. Latent feature evaluation via Univariate Cox-PH ($p < 0.05$).
4. K-means clustering (Silhouette optimized) to define High/Low Risk groups.

Phase B: Supervised Biomarker Extraction

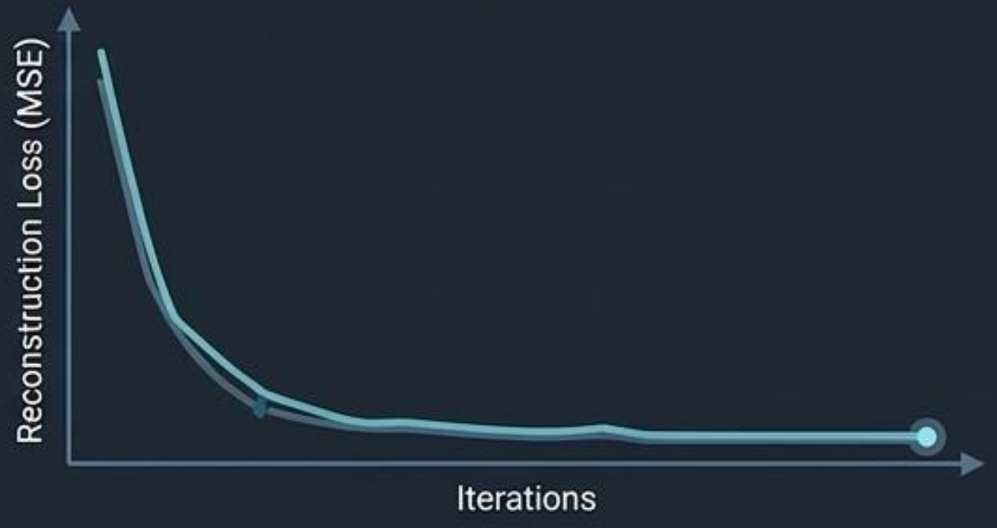
1. Labeling raw data with Phase A predicted risk clusters.
2. Likelihood-ratio feature selection.
3. C5 rule-based decision tree classification.
4. Biomarker identification & experimental database cross-validation.

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Unlocking Peak Precision: The HPOAE Bottleneck



The Optimization Plot



Optimization (Scikit-Optimize):

- Tested permutations of ReLu, Sigmoid, Linear, Tanh against various optimizers.

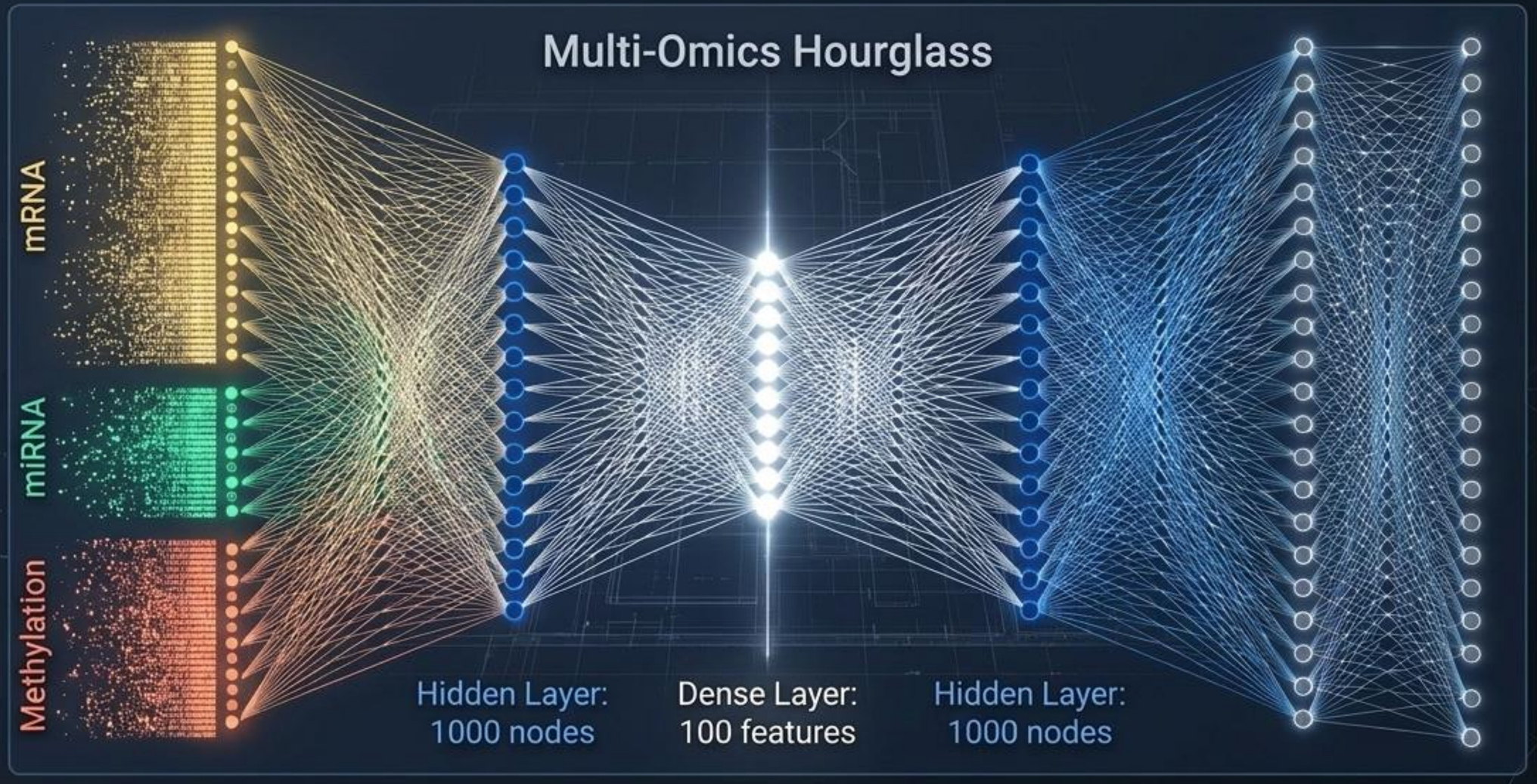
Winning Parameters: ReLU activation, RMSprop optimizer, Batch size 32.

Reconstruction Loss: Minimized to MSE = 0.015.

Note: Strong cluster consistency established (Kuder–Richardson criterion = 0.912).

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The Hyper-Parameter Optimized Autoencoder (HPOAE)



By forcing massive biological arrays through a 100-node bottleneck, the unsupervised neural network compresses chaos into a pure, low-dimensional representation of patient risk.

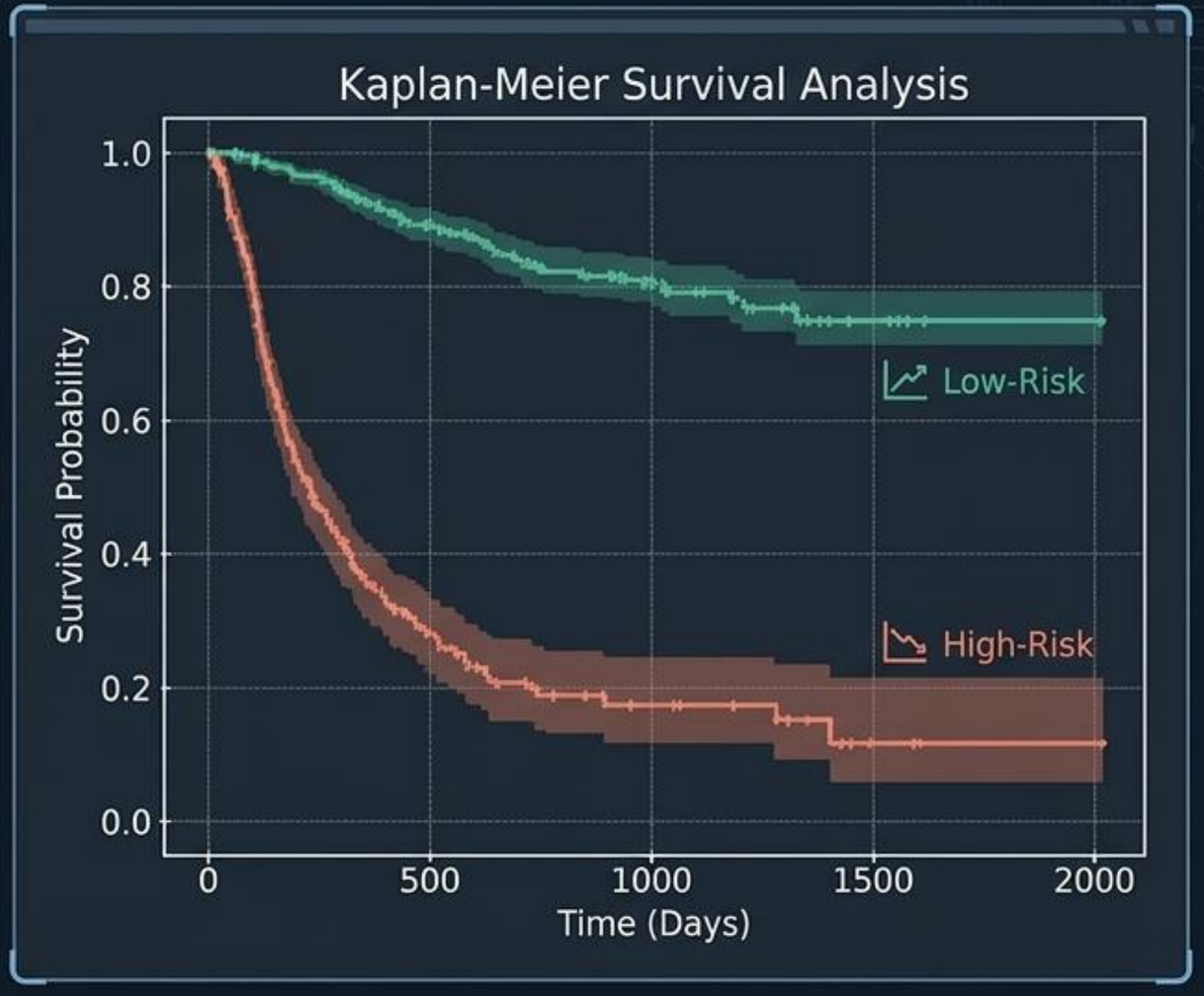
Translating Latent Nodes to Clinical Risk Groups

Filtering Mechanism

100 bottleneck features filtered through Univariate Cox Proportional Hazards model (log-rank $p < 0.05$), isolating 80 survival-associated features.

Patient Segmentation

K-means clustering applied to the 80 features. Silhouette index identified k=2 as the optimal cluster number.



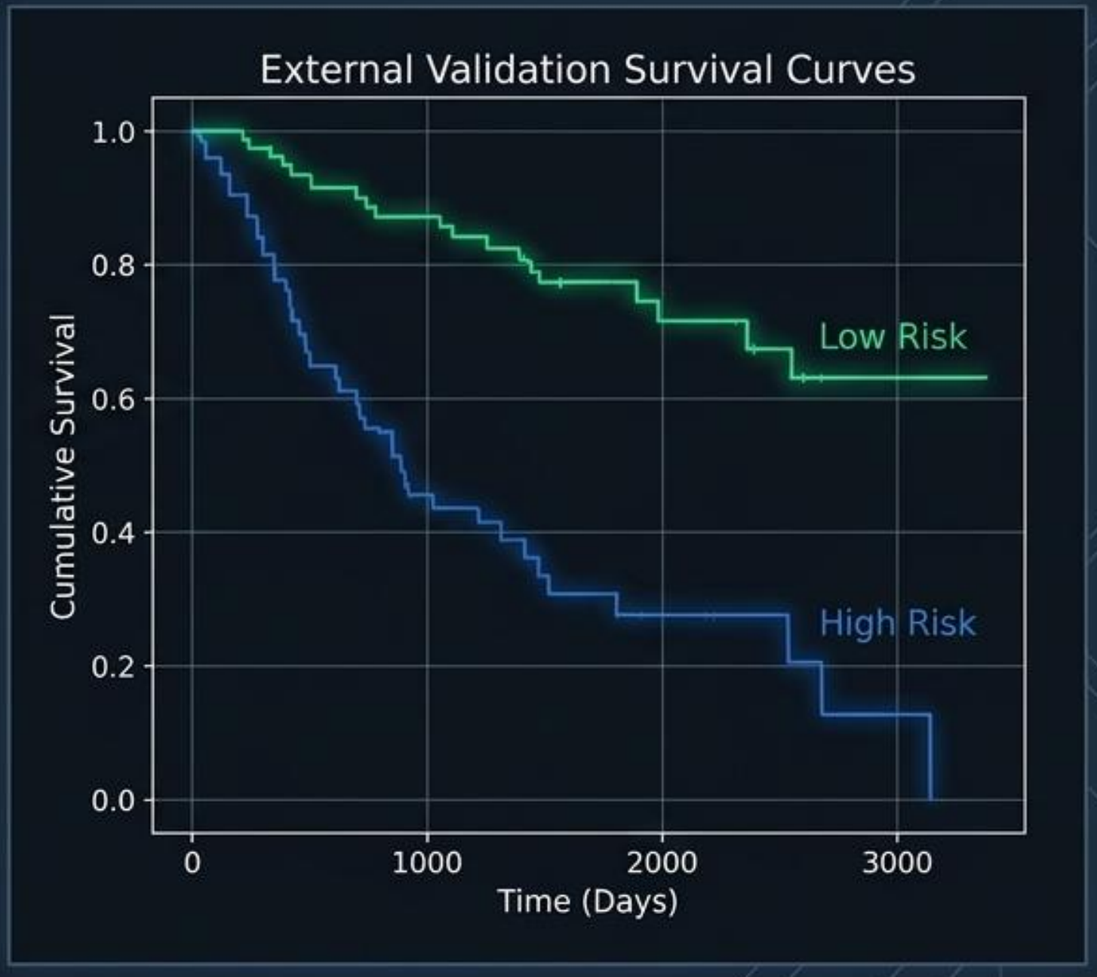
Result: A statistically significant, stage-independent separation of the cohort into two distinct clinical trajectories.

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Generalizable and Robust: External Validation

- **Cohort details:** E-GEOD-17538 Dataset (n=232 samples).
- **Process:** Fed into the exact same deep autoencoder and Cox-PH model trained exclusively on the TCGA cohort.

Bottom Line: The model is highly resistant to over-fitting, proving robust against stochastic variations and independent datasets.



The Model Showdown: HPOAE vs. Baseline Frameworks

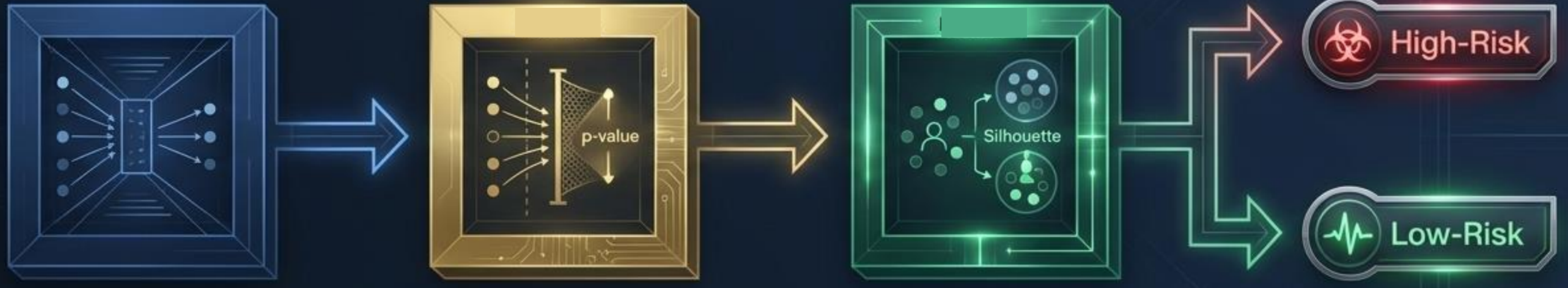
Precisiion Comparison Matrix			
Framework	**Log-Rank**	**Breslow**	**Tarone-Ware**
Optimized Autoencoder	14.272	12.012	13.131 (Sig: 0.001)
Penalized PCA (PPCA)	9.914	2.105	5.761 (Sig: 0.002)
Normal Autoencoder	8.375	2.743	4.490 (Sig: 0.004)

⚠ Amber Flag - Reporting Discrepancy: The manuscript text and abstract repeatedly claim a Breslow statistic of 13.13 ± 1. However, Table 2 reports 12.012 for Breslow and 13.131 for Tarone-Ware.

Note: These statistics indicate the significance of survival group separation, not direct predictive accuracy for individual patients.

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Translating Hidden Nodes to Clinical Risk



1. The Bottleneck Representation

100 abstract deep-learning features.

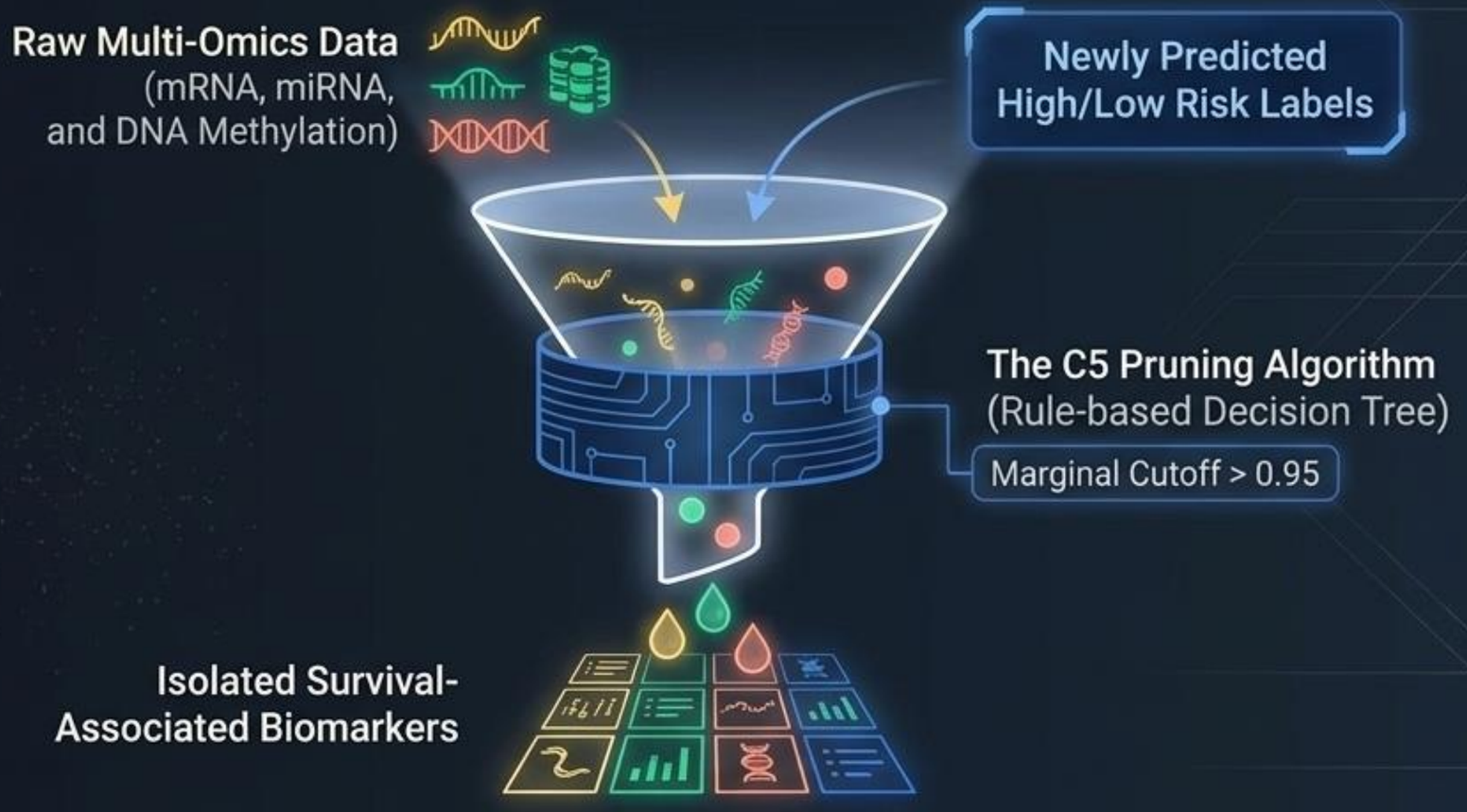
2. Univariate Cox-PH Model

Filtering for survival significance (log-rank $p < 0.05$) isolates 80 associated features.

3. K-Means Segmentation

Clustering patients ($k=2$) optimized by the Silhouette index.

Phase II: Extracting the Biological Rules



The deep learning model acted as a ‘black box’ to find the risk groups. We then deployed a supervised C5 Decision Tree to force the data to explain itself, extracting explicit conditional rules.

The Multi-Omics Advantage: The Whole > The Parts

The Integration Advantage Matrix (10-fold CV Training)

mRNA Only	F1: 0.95	Accuracy: 0.97
miRNA Only	F1: 0.94	Accuracy: 0.96
Methylation Only	F1: 0.91	Accuracy: 0.94
Multi-Omics (3-Omics)	Precision: 0.978	F1: 0.9375 Accuracy: 0.969

While single omics look high on training, the multi-omics framework yields the most cross-validated precision and stability across the entire architectural model, capturing complex epigenetic interactions single-omics completely miss.

The Survival Biomarker Dashboard

28 mRNAs
ACTL10
CAMK2B
CCL24
LBP
ZMYM1

10 miRNAs
hsa-miR-485-5p
hsa-miR-33b-3p
hsa-miR-34a-5p

11 Methylated Genes
ALG13
ANGPT4
E2F2
GPR182

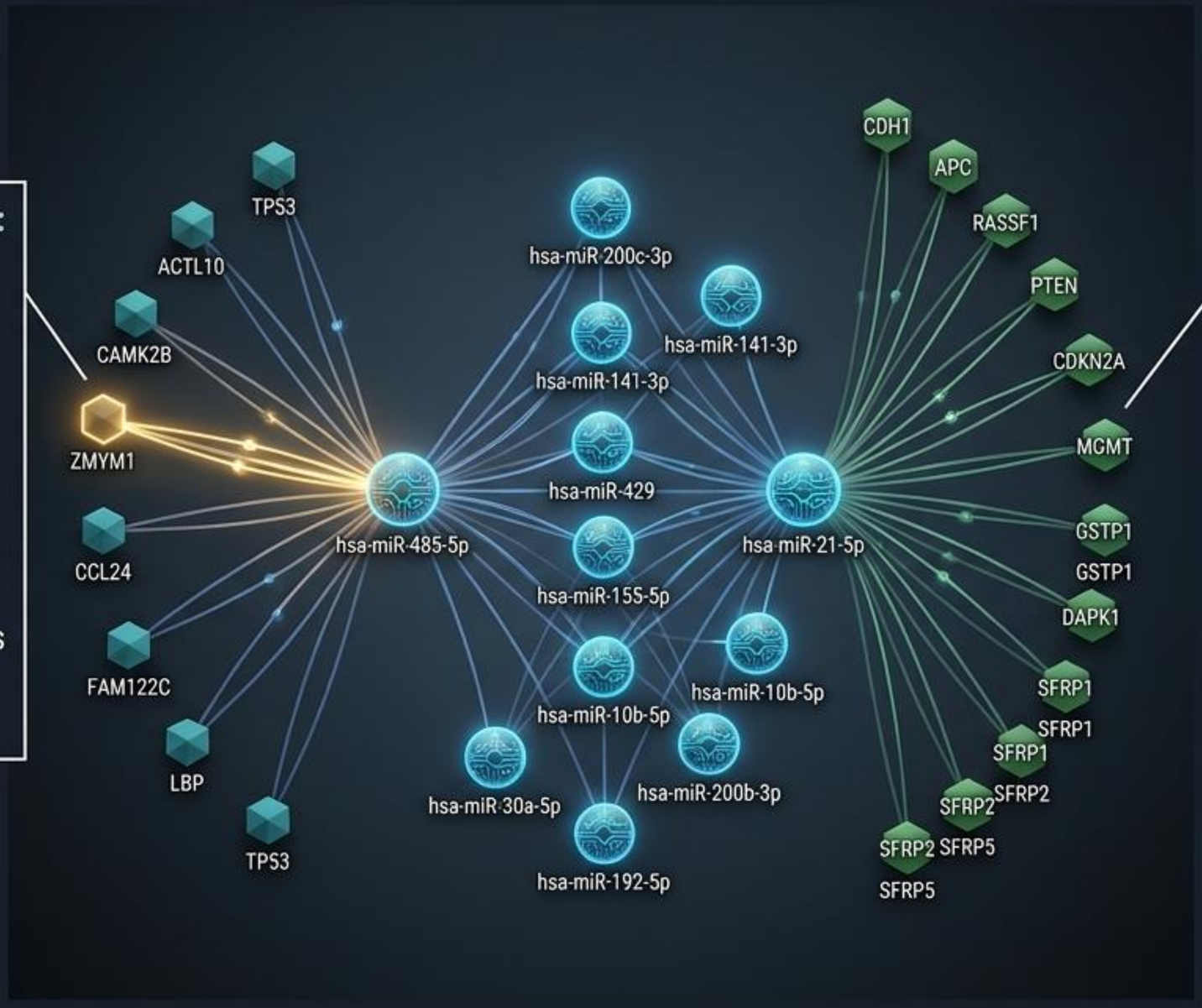
Feature selection isolated these exact biomarkers from >42,000 features with an importance marginal cutoff ≥ 0.95.

The Epigenetic Network: Validation & Interpretation

The ZMYM1 & miR-485-5p Axis:

The computational model blindly selected ZMYM1 and its regulator hsa-miR-485-5p.

Database Validation:
Experimental databases (mirTarBase) confirm miR-485-5p directly targets ZMYM1 (and tp53), linking these mathematically derived features to actual cancer metastasis mechanisms (EMT).



Key Insight Distinction:
6 identified mRNAs were predicted by TargetScan/miRDB (ACTL10, CAMK2B, CCL24, FAM122C, LBP, ZMYM1). Only a subset possesses direct experimental validation in independent biological literature; the rest serve as novel computational candidate biomarkers.

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Biological Validation: The ZMYM1 & miR-485-5p Axis

The Discovery

The algorithm blindly identified ZMYM1 (zinc finger MYM-type containing 1) and its regulator hsa-miR-485-5p as critical survival determinants.



The Validation

External experimental databases confirm miR-485-5p directly targets ZMYM1 (and tp53), linking these mathematically derived features to actual cancer metastasis mechanisms (EMT).

Scientific Scrutiny: Strengths and Methodological Limitations

Strengths & Innovations:

- Successful hyperparameter optimization, dramatically lowering MSE vs. standard PCA.
- Robust clustering achieved independent external confirmation (E-GEOD-17538).
- Excellent translation of unsupervised clusters into interpretable C5 biological rules.

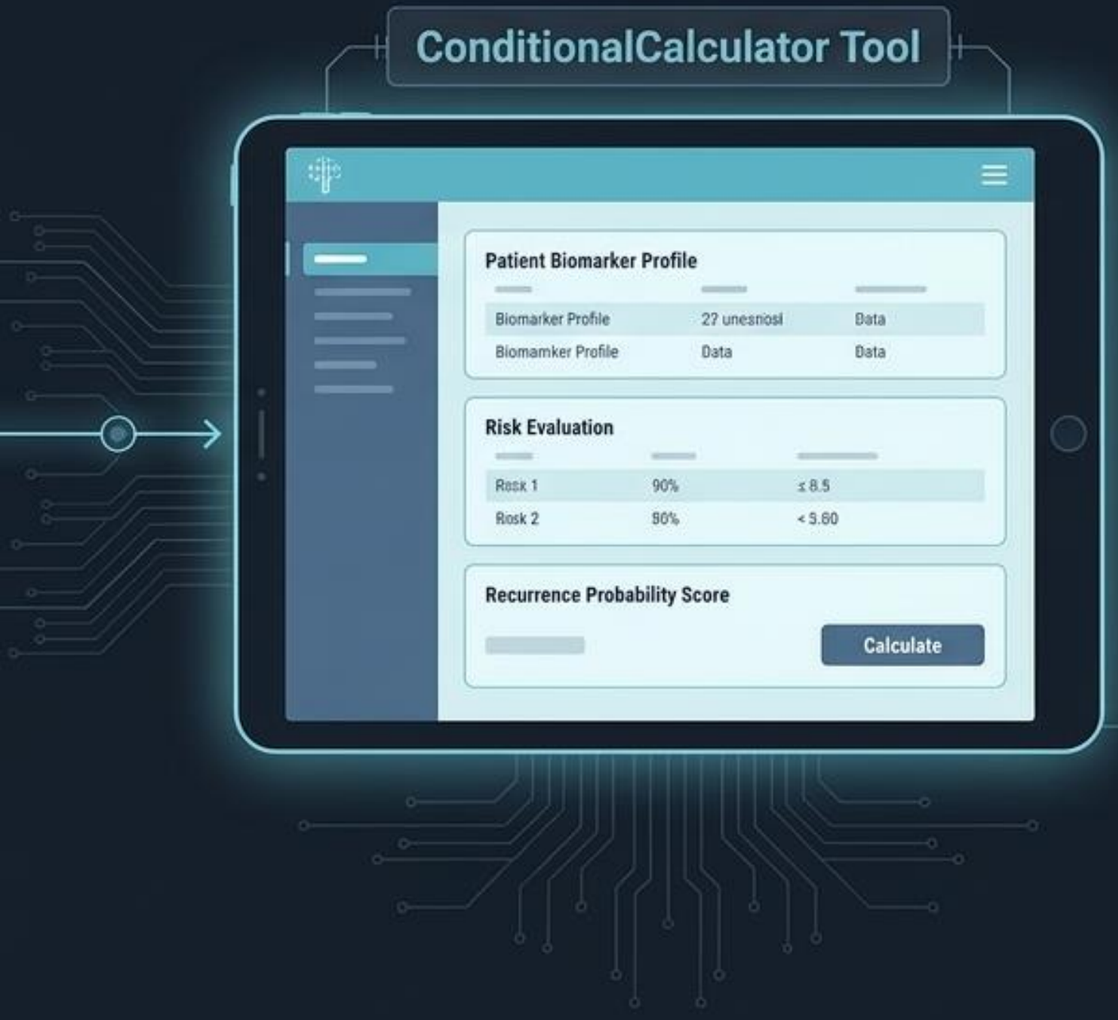


Limitations & Reporting Discrepancies:

- Overfitting Evidence: Massive drop in 3-omics metrics from Training (0.97) to Test (0.56).
- The Multi-Omics Premise: Methylation-only test data proved superior to integrated data.
- Sample Size vs. Dimension: 211 intersecting samples is severely underpowered for >37k input features.
- Data Contradictions: Title claims clinical data integration, but methods show HPOAE ran strictly on omics data; abstract statistical typos (Breslow 13.13 vs 12.01).

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From Computational Theory to Clinical Translation



The ConditionalCalculator Tool: The authors embedded the C5 D-tree rules into an open-access platform.

Utility: Allows evaluation of a patient's risk profile using only the distilled candidate biomarkers rather than requiring 42,000-feature sequencing.

Future Research Imperatives:

1. **Cohort Scaling:** Testing the framework on massive, thousands-strong diverse cohorts.
2. **True Clinical Integration:** Actually feeding patient clinical covariates (age, stage, treatment history) into the deep learning bottleneck alongside omics data.
3. **Wet-Lab Validation:** Prospective, direct experimental validation of the novel miRNA-mRNA target pairs (e.g., ACTL10 and LBP).

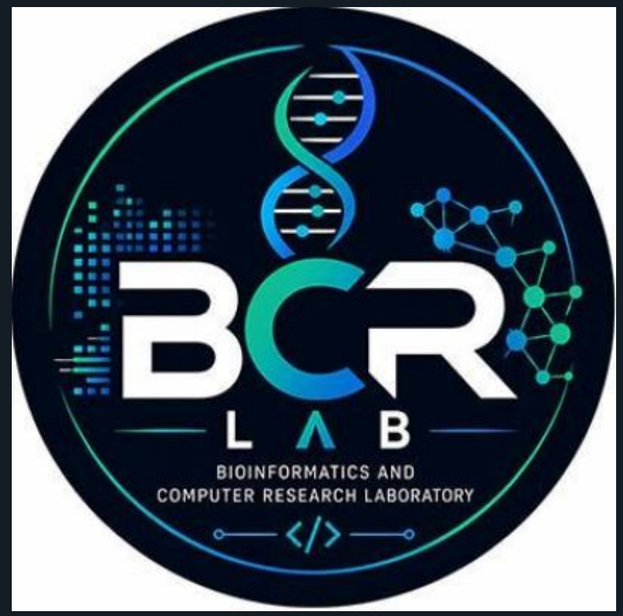
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siamaksalimy

siamaksalimy

SalimySiamak

Salimy_BIO



BCRLAB.IR

info@BCRLAB.IR


Thanks